

Active Directory Home Lab

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Project Overview

Ran a Windows Server 2022 Active Directory environment in VirtualBox consisting of one Domain Controller and two Windows 10 clients. The lab was used to practice user management, Group Policy, permissions, troubleshooting, and common administrative tasks.

Lab Architecture

- Domain Controller: Windows Server 2022
 - Client1: Windows 10
 - Client2: Windows 10
 - Domain: adlab.local
 - DNS hosted on Domain Controller
 - Internal VirtualBox network
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Features Implemented

Domain Infrastructure

- Created AD forest: adlab.local
- Configured DNS
- Joined Windows clients to the domain

User & Group Management

- Created Organizational Units
- Created users and groups
- Disabled and re-enabled accounts
- Reset passwords
- Forced password changes

Group Policy

- Account lockout policy
- Desktop wallpaper deployment

- Disable Control Panel
- Disable USB storage devices
- Network drive mapping

File Sharing & Permissions

- Shared HR folder
- Configured NTFS permissions
- Configured group-based access control

Remote Administration

- Configured Remote Desktop
 - Connected to domain workstations remotely
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Troubleshooting Exercises

DNS Failure

Changed workstation DNS settings to an incorrect server and saw:

- Domain name resolution failed
- Access by IP address still worked
- Domain resources became unavailable

Changed DNS settings back to normal and verified normal operation.

Group Policy Troubleshooting

USB restriction policy failed to apply.

Discovered that the GPO was Linked to a OU that only had user accounts.

Moved workstations into a dedicated OU and linked the GPO to that OU and USB restriction policy worked.

Network Connectivity Failure

Disabled the workstation network adapter and compared symptoms against the DNS failure scenario.

Observed complete loss of connectivity rather than name-resolution-only failures.

Key Takeaways

- Active Directory relies heavily on DNS functionality.
- User and Computer GPOs apply differently.

- Existing user sessions may remain active after account changes.
 - Group membership changes do not always take effect until reauthentication.
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Technologies Used

- Active Directory
- DNS
- Group Policy
- Windows Server 2022
- Windows 10
- VirtualBox
- SMB File Sharing
- Remote Desktop